

Problem 1

$x = \text{"ILOVEDISCRETEMATHS"}$

$m_x = \text{ILOVEDISCRETEMATHSAB}$

$l = 20$

$$p = \frac{l}{k} = \frac{20}{5} = 4$$

$$A_x = \begin{pmatrix} I & D & E & T \\ L & I & T & H \\ O & S & E & S \\ V & C & M & A \\ E & R & A & B \end{pmatrix}$$

Cypher Blocks = (IDET, LITH, OSES, VCMA, ERAB)

Cypher Text: $C = \text{IDETLITHOSESVCMAERAB} \times$

Need padding?

$$|x| = 18 \quad k = 5$$

$$\begin{aligned} \text{Remainder} &= |x| \bmod k \\ &= 18 \bmod 5 \\ &= 3 \end{aligned}$$

$$k - \text{Remainder} = 5 - 3$$

$$= 2$$

∴ Need to pad 2 characters.

$y = \text{ILOVEDISCRETEMATHSAB}$

$m_y = y$

$l = 20$

$$p = \frac{l}{k} = \frac{20}{5} = 4$$

$$A_y = \begin{pmatrix} I & D & E & T \\ L & I & T & H \\ O & S & E & S \\ V & C & M & A \\ E & R & A & B \end{pmatrix}$$

Cypher Blocks for $y = (\text{IDET}, \text{LITH}, \text{OSES}, \text{VCMA}, \text{ERAB})$

Cypher Text for y : $C = \text{IDETLITHOSESVCMAERAB} \times$

Need Padding?

$$|y| = 20 \quad k = 5$$

$$\begin{aligned} \text{Remainder} &= |y| \bmod k \\ &= 20 \bmod 5 \\ &= 0 \end{aligned}$$

Since there is no remainder,
no need padding.

Problem 2

- Because two or more different messages (m) can produce the same cypher text, as shown in problem 1.
- For instance $x = \text{ILOVEDISCRETEMATHI}$ and $y = \text{ILOVEDISCRETEMATHSAB}$ both produce the same cypher text of $\text{IDETLITHOSESVCMAERAB}$.
- This means that the function would not be injective. It will not be a one-to-one function, as one element in the codomain (cypher text) can be mapped to at least one or more element in the domain (the plain text message).
- As the encryption function is not injective, it cannot be used as an encryption function for a cryptosystem.

Problem 3

1. $M \in (\Sigma \cup \{Y\})^*$ The message space M is the set of all strings of English alphabet and the padding character.

$C = M$ The cypher space C is the set of all string of the English alphabet and the padding character.

$K \in \mathbb{N}$ The key space K is the set of all natural numbers.

2. Definition: $d_k: C \rightarrow M$ given $c \in C, k \in K$

- Let l be the length of cypher text C , such that $l = |C|$. Now, let p be the message block size, such that $p = \frac{l}{k}$.
- Create a $p \times k$ array A , where the first column of A contains the string $c_1, c_2, \dots, c_p$ and the second column contains the string $c_{p+1}, c_{p+2}, \dots, c_{2p}$ and so on. The string is written from top to bottom.

$$A = \begin{pmatrix} c_1 & c_{p+1} & \dots & c_{(k-1)p+1} \\ c_2 & c_{p+2} & \dots & c_{(k-1)p+2} \\ \vdots & \vdots & \ddots & \vdots \\ c_p & c_{2p} & \dots & c_{kp} \end{pmatrix}$$

- To obtain the message m , let m_i be a message block where each block is a string obtained by reading row i from left to right. The original message m will be obtained by concatenating the message blocks in numerical order, that is $m = m_1 m_2 \dots m_p$.

Problem 3 (continued)

- If the padding character γ occurs in m , let z be the index of the padding character γ in m .

The original text t is obtained by obtaining a substring of m from m_1 to $m_{(z-1)}$ such that $t = m_1 m_2 \dots m_{(z-1)}$.

- Else, if the padding character γ does not occur in m , we let text message $t = m$.

- Hence, for any $t \in M$, $d_k(e_k(t)) = t$ $\times$

3.

- Yes, fixed row column cypher V2 (M, C, K, e, d) is a valid cryptosystem as for all $k \in K$, meets the following requirements.
- For $k \in K$, e_k perfectly maps between the message space M and the cypher space C . This is said as the special padding character is used for the padding system, so two different plaintext cannot map to the same cypher text, making this a bijection, where given a unique key, all messages maps to one unique cypher text.
- As e_k is a bijection, this allows for an inverse to occur, where $d_k = e_k^{-1}$, as it can reverse the bijective mapping, allowing us to obtain the original plain text t given the cypher text c and the key k , $d_k(e_k(t)) = t$.
- Hence, fixed row column cypher V2 (M, C, K, e, d) satisfies the definition of a cryptosystem.

Problem 4

$x = \text{ILOVECOMPUTERS}$

$s = \text{length of } x = 14$

$k = (4, p)$

$p = (3, 1, 2, 4)$

$$\frac{s}{k} = \frac{14}{4} = 3.5 \Rightarrow k \text{ cannot divide } s$$

∴ Padding required

$$k - (s \bmod k)$$

$$= 4 - (14 \bmod 4)$$

$$= 2 \rightarrow \text{padding with 2 characters, let } y = !$$

Thus, $m = \text{ILOVECOMPUTERS!A}$, $l = 16$

$$p = \frac{l}{k} = \frac{16}{4} = 4$$

$$m = b_1 b_2 b_3 b_4 = (\text{ILOV, ECOM, PUTE, RS!A})$$

$$c = b_{p(1)} b_{p(2)} b_{p(3)} b_{p(4)}$$

$$c = b_3 b_1 b_2 b_4$$

$$c = \text{PUTEILOVECOMRS!A} *$$

$y = \text{IEPRLCUSOOT!VMEA}$ $s = \text{length of } y = 16$ $k = (4, p)$ $p = (3, 1, 2, 4)$

$$\frac{s}{k} = \frac{16}{4} = 4 \Rightarrow k \text{ can divide } s$$

∴ padding not required

$$m = y \quad l = s$$

$$p = \frac{l}{k} = \frac{16}{4} = 4$$

$$m = b_1 b_2 b_3 b_4 = (\text{IEPR, LCUS, OOT!, VMEA})$$

$$c = b_{p(1)} b_{p(2)} b_{p(3)} b_{p(4)}$$

$$c = b_3 b_1 b_2 b_4$$

$$c = \text{OOT!IEPRLCUSVMEA} *$$

Problem 5

Yes, the block permutation cypher can be used as a valid cryptosystem. Furthermore, encrypting a message m by first applying fixed row column cypher V2 and then block permutation cypher is also a valid cryptosystem.

1. Block Permutation Cypher

- Given a fixed key (k, P) , the block permutation (BP) cypher produces a bijective mapping between the message space and cypher space. Since it uses the fixed row column (FRC) cypher V2 padding system, it is also invertible to remove the padding. Furthermore, the permutation function P is also a bijection, which allows for decryption to occur by passing P^{-1} to the obtain back the original message. $d^{BP}(k, P) = e^{BP}(k, P^{-1})$

2. Composite Cryptosystem - BP Cypher and FRC V2 Cypher.

- It is established earlier that FRC V2 and BP are valid cryptosystems and are both bijective.
- Thus, a composition of bijective functions between FRC V2 cypher and BP cypher remains a bijection, which means the composite function has an inverse, therefore a decryption exist.

$$e_{(k, P)}^{\text{composite}} = e_{(k, P)}^{BP} \circ e_{(k)}^{FRC V2} : M \rightarrow C$$

- The decryption:

$$d_{(k, P)}^{\text{composite}} = d_{(k)}^{FRC V2} \circ d_{(k, P)}^{BP} : C \rightarrow M$$

- Bijections are injections and surjection:

- Injection occurs here as there is a one-to-one relationship between the message space and cypher space.

- Surjection occurs as every cypher text is mapped to one element in the message space.

- Hence, the composition function between BP cypher and FRC V2 is valid.

Problem 6

Case 1: $k = 1$

If $k = 1$, putting the message of length l into a grid will result in one single row. This turns the cypher $c = b_i$, where $b_i = t$, so $c = t$. Hence, the key is poor when $k = 1$.

Case 2: $k \geq l$

When k is $\geq l$, the message length will always be equal to k , as padding is needed until $p = l$, where $p = \frac{l}{k}$. This will result in the grid having one column, with k rows. $c = b_1, b_2, \dots, b_k$. But since each row only contains one character, $c = t!x$, where $x = \Sigma^*$ and is the padding. Hence, when $k \geq l$, it is a poor key.

Case 3: $1 < k < l$

When $1 < k < l$, it will not result in a single row or single column grid. This ensures that $c \neq t$ or $c \neq t!x$, where $x = \Sigma^*$. Hence, when $1 < k < l$ the key is not poor.

∴ Total Not Poor keys = $l - \text{case 1} - \text{case 2}$
for plain text of length l

$$= \underline{\underline{l - 2}}$$

Problem 7

Given t with Length l $l=8$.

Since cypher text result in length 8, no padding will be used.

Since, $P = \frac{l}{k}$, l must be divisible by k .

$$\therefore l=8, k=\{1, 2, 4, 8\}$$

Since $|P| = k$, possible combinations of P is $k!$.

e.g. when $k=4$, $|P| = k! = 4! = 24$

For every k , there is one combination of P that results in a poor key, where $C=t$.

i.e., when the permutation function is an identity permutation, where the order does not move any elements.

Poor keys: $k=1, P=(1)$

$$k=2, P=(1, 2)$$

$$k=4, P=(1, 2, 3, 4)$$

$$k=8, P=(1, 2, 3, 4, 5, 6, 7, 8)$$

$$\text{Total non-poor keys} = (1! - 1) + (2! - 1) + (4! - 1) + (8! - 1)$$

$$= 0 + 1 + 23 + 40319$$

$$= \underline{\underline{40343}}$$

Problem 8

- For encrypting plain text with length λ and resulting in a cypher text in length λ also, no padding must occur.
- A set of possible keys Z can be created such that no padding is required for the text, that is: $Z = \{k \in \mathbb{N} \mid \lambda \bmod k = 0\}$
- All elements of Z are keys that can divide λ .

Poor Keys:

- Poor key occurs when the combination P is an identity permutation, where it does not move any elements after being applied.
- For every k in set Z , there will be one identity permutation.
- e.g. $(k=4, P=(1,2,3,4))$, $(k=2, P=(1,2))$

Obtain number of non-poor keys:

- The total number of non-poor keys for plain text with length λ is obtained by the sum of the k_i factorial minus 1, for $k_i \in Z$.

$$\sum_{i=1}^{|Z|} (k_i! - 1), \quad k_i \in Z$$

Problem 8 (continued)

- For every key k_i in set Z , the total possible key set (k_i, P) is given by $k_i!$, as the permutation function has k_i elements. The total permutation for k_i elements is $k_i!$.
- Since each key has one poor key pair. It can be said that the number of non-poor keys is $(k_i! - 1)$.
- The sum of all non-poor keys for all k in set Z will be the answer.

$$\sum_{i=1}^{|Z|} (k_i! - 1) \quad \# \text{, such that } k_i \in Z$$